



ELECTRIC CURRENT — FULL MIND MAP

विद्युत धारा · Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Current, Potential, Resistance, Power & Electrical Energy
2. Dynamo, Motor, Battery & Energy Conversion
3. Bulbs, Lightning, Earthing, Fuse & Household Safety
4. CFL, LED, Fluorescent Tube & Lighting Technology
5. Potential, Magnetic Field & Transformer
6. AC–DC Generation, Rectifier & Power Use
7. Earth Magnetism, Solar Power & Distributed Energy
8. Pumped Storage & Wind Power

1. Current, Potential, Resistance, Power & Electrical Energy

- Current density $J = I/A$ hota hai aur direction conventional positive-charge flow ki hoti hai; isliye current density vector quantity hai. Relative dielectric constant $\epsilon_r = \epsilon/\epsilon_0$ dimensionless hota hai. [Q1–Q2]
- Charge q ko potential difference V se accelerate karne par gained energy $E = qV$. Electron ke liye magnitude eV joule hota hai. [Q3]
- Electrical energy: $E = P \times t$. 1 unit = 1 kWh = 3.6×10^6 J. Isliye $100 \text{ W} \times 10 \text{ h} = 1$ unit; $15 \times 200 \text{ W} \times 24 \text{ h} = 72$ units; $60 \text{ W} \times 5 \text{ h} \times 30 \text{ days} = 9$ units. [Q4–Q11]
- Power relation $P = VI$. Ohm law se $R = V/I$. Household parallel circuit me same voltage par $P = V^2/R$; dimmer bulb ka resistance larger hota hai, aur identical bulbs parallel me series se brighter hote hain. [Q12–Q16]
- Parallel resistors ke liye $1/R = 1/R_1 + 1/R_2$. Resistor colour code brown-green-black = 15Ω ; 15 V par current 1 A. Capacitive reactance $X_c = 1/(2\pi fC)$, so frequency badhne par X_c ghatta hai. [Q17–Q19]

2. Dynamo, Motor, Battery & Energy Conversion

- Dynamo/generator mechanical energy ko electrical energy me convert karta hai; turbine mechanical rotation provide kar sakta hai. Electric motor ulta conversion karta hai: electrical $\rightarrow$ mechanical. [Q20–Q26]
- Dynamo electromagnetic induction par based hota hai. Battery electrochemical cells ke through stored chemical energy ko electrical energy me convert karti hai. [Q27–Q28]
- Low-voltage operation me same required power/torque ke liye motor higher current draw kar sakta hai, causing I^2R heating. Faraday constant $F \approx 96485 \text{ C/mol}$ electrons ek universal constant hai. [Q29–Q30]
- Typical car lead-acid battery $\approx 12 \text{ V}$ hoti hai; electrolyte H_2SO_4 aur electrodes Pb/PbO_2 hote hain; capacity ampere-hour me. Ni-Cd batteries rechargeable applications me use hoti hain. [Q31–Q32]



EXAM TRAP — Q24 & Q27 — Idealisation / Principle Wording

- ▶ Q24 ka “no heat loss” ideal theoretical conversion ke context me hai; real motors me resistance, friction etc. se losses hote hain.
- ▶ Standard principle: generator/dynamo $\rightarrow$ electromagnetic induction; motor ka primary working principle current-carrying conductor par magnetic force/torque hai. Source Q27 me “Also D” note ko universal rule na samjhein.

3. Bulbs, Lightning, Earthing, Fuse & Household Safety

- Incandescent bulb filament tungsten ka hota hai because very high melting point; operating filament temperature roughly $2000\text{--}2500^\circ\text{C}$ range me hota hai. [Q33–Q37]
- Lightning me enormous electrical energy hoti hai. Lightning conductor charge ko low-resistance path se ground tak le jaakar building protect karta hai. [Q38–Q39]
- Three-pin plug ka longest/thick earth pin ground terminal se connected hota hai; earthing shock-protection ke liye leakage current ko ground path deti hai. Same potential par connected bodies ke beech net current flow nahi hota. [Q40–Q42]
- Fuse safety device hai: heating effect I^2R par kaam karta hai; low melting point aur relatively high resistivity/resistance material excess current par melt karke circuit break karta hai. [Q43–Q46]
- Current-carrying metal wire ka temperature resistance losses se rise karta hai; household wiring fundamentally parallel hoti hai. Black insulation older conventions me neutral wire ko denote kar sakti hai; dry human-body resistance order $\approx 10^4 \Omega$ source value hai. [Q48–Q51]
- Source Q47 halogen lamp filament ko “tungsten + sodium alloy” mark karta hai. [Q47]

⚠ EXAM TRAP — Q36 & Q47 — Source-specific Bulb Statements

- ▶ Ordinary incandescent lamp life ka dominant mechanism tungsten filament evaporation/thinning and hot-spot failure hota hai; source Q36 ke all-three reasons ko source-answer ke roop me preserve kiya gaya hai.
- ▶ Halogen lamp me filament normally tungsten hota hai aur bulb me halogen gas (iodine/bromine compounds) use hoti hai. "Tungsten + sodium alloy" source Q47 ko standard fact na samjhein.

4. CFL, LED, Fluorescent Tube & Lighting Technology

- CFL = Compact Fluorescent Lamp. LED electroluminescence principle par semiconductor junction se light produce karta hai. LED generally CFL se longer-life aur more energy-efficient hota hai. [Q52–Q55]
- Street-light comparison: sodium lamps nearly monochromatic yellow output aur broad emission de sakte hain; LEDs directional, colour-controllable aur typically longer-life technology hain. Source Q56 correct set 1 & 3 deta hai. [Q56]
- Fluorescent tube low-pressure mercury vapour discharge + phosphor coating use karti hai; electrical discharge UV generate karta hai aur phosphor visible white light emit karta hai. Source gas option "mercury and neon" diya hai. [Q57–Q59]
- Incandescent bulbs me inert gas mixture such as argon/nitrogen commonly hota hai. Fluorescent choke starting ke time high-voltage pulse provide karta hai aur operation me current limit karta hai. [Q60–Q61]

⚠ EXAM TRAP — Q59–Q61 — Tube-light Mechanism

- ▶ White light ka correct physical mechanism phosphor fluorescence hai; "molecules oscillating" source option ko literal mechanism na samjhein.
- ▶ Choke sirf "step-up transformer" nahi hai: fluorescent circuit me inductive ballast starting pulse aur current limiting dono roles karta hai.

5. Potential, Magnetic Field & Transformer

- Electric potential scalar quantity hai. $P = VI$, so V aur I dono double hon to power $4\times$ ho jati hai. Wire resistance $R = \rho L/A$; given length, diameter and resistivity ratios Q64 me both wires ka R equal nikalta hai. [Q62–Q64]
- Same-radius charged drops coalesce karen: total charge nQ , new radius $n^{1/3}r$, therefore potential $V_{\text{new}} = n^{2/3}V$. [Q65]
- N -turn circular coil centre field $B = \mu_0 NI/(2r)$; $N=100$, $r=0.1$ m, $I=1$ A gives 6.28×10^{-4} T. [Q66]
- Transformer AC voltage ko step-up/step-down karta hai and electromagnetic induction par works; steady DC par normal transformer action nahi hota. Mobile charger front-end mains voltage ko lower suitable level me convert karta hai. [Q67–Q70]

6. AC–DC Generation, Rectifier & Power Use

- Correct energy conversions: generator mechanical→electrical; motor electrical→mechanical. AC generator/dynamo alternating current produce kar sakta hai. [Q71–Q72]
- Basic generator distinction: AC generator uses slip rings; DC generator uses split-ring commutator. Storage battery charging ke liye DC required hota hai, isliye raw AC directly preferable nahi. [Q73–Q74]
- Rectifier AC ko DC me convert karta hai. India me electricity consumption kWh me billed hoti hai aur grid frequency 50 Hz hoti hai, 60 Hz nahi. [Q75–Q77]
- Electric traction diesel pollution reduce kar sakti hai; source argument set me "diesel pollutes" strong argument hai. Typical appliance power ascending: fan < television < electric iron/press < electric kettle. [Q78–Q79]

7. Earth Magnetism, Solar Power & Distributed Energy

- Earth ka magnetic field conducting liquid outer core me circulating electric currents / geodynamo action se generated maana jata hai. [Q80]
- Artificial satellites me solar/photovoltaic cells sunlight ko electrical energy me convert karte hain. Space-based solar power concept orbit me collection aur Earth par wireless transmission propose karta hai. [Q81–Q82]
- Photovoltaic technology light ko directly DC electricity me convert karti hai; solar thermal pehle heat generate karta hai, phir heat-engine route se electricity. [Q83]
- Solar power surface/submersible aur centrifugal/positive-displacement pumps dono chala sakti hai. Most common solar cells silicon semiconductor based hote hain. [Q84–Q85]

- Photovoltaic cell = solar cell; photovoltaic/photoelectric devices light energy ko electrical energy me convert karte hain and satellites/digital systems me widely used hain. [Q86–Q90]
- Net metering rooftop solar owner ko grid me export ki gayi electricity ka billing credit deta hai. Solar tariff/manufacturing claims source Q92 me both incorrect marked hain. [Q91–Q92]
- Distributed Energy Resources (DERs) me rooftop PV, battery storage, biomass generators aur fuel cells jaise small/local resources include hote hain. [Q93]



EXAM TRAP — Q82 & Q92 — Policy/Data Claims Can Age

- ▶ SBSP “99% uninterrupted” aur national-goal wording source/exam context ka statement hai, general engineering guarantee nahi.
- ▶ Solar manufacturing rankings/tariffs time-sensitive policy facts hain; source answer ko question-year context me padhein.

8. Pumped Storage & Wind Power

- Pumped-storage hydropower do different-elevation reservoirs ke beech water move karke energy store/release karta hai; essentially long-duration grid energy storage. [Q94]
- Source Q95 India ko wind installed capacity me world rank 4 aur Gujarat ko high wind potential state batata hai; “400 GW installed” statement false marked hai. [Q95]
- Source Q96 (question-time data) me Madhya Pradesh ko third-ranked wind-power state batana incorrect marked hai; Gujarat/Tamil Nadu leading states ke roop me diye gaye hain. [Q96]



EXAM TRAP — Q95–Q96 — Wind Capacity Figures Are Time-sensitive

- ▶ Installed capacity, state shares aur rankings year-to-year change hote hain. In questions ke numerical/ranking claims ko source-question date ke context me retain karein.

✓ Source coverage: Q1–Q96